

AI-AUGMENTED BACKEND DEVELOPMENT

AI BlogNest API

8. Project Demonstration Phase

Demo preparation, execution flow, AI demonstration, and submission evidence

1. Demonstration Objective

The demonstration phase presents the working AI BlogNest API from setup through successful API execution. The sample's final evidence sequence includes server execution, API testing, user registration, login, blog creation, fetching blogs, AI blog generation, summarization, demo video, API testing video, and code reference. ■filecite■turn0file0■L189-L205■

2. Demonstration Preparation

- Start MongoDB and verify database connectivity.
- Start the Node.js backend using npm start.
- Open Postman or Thunder Client.
- Load the API collection or enter the required endpoints.
- Prepare valid test credentials and sample blog inputs.
- Prepare an AI topic for generation and a blog article for summarization.

3. Demonstration Flow

- Step 1 – Show the project structure and server configuration.
- Step 2 – Demonstrate user registration.
- Step 3 – Demonstrate user login and JWT authentication.
- Step 4 – Demonstrate a protected API request.
- Step 5 – Create a blog post.
- Step 6 – Fetch the created blog.
- Step 7 – Demonstrate role-based access behavior.
- Step 8 – Send a topic to the AI blog-generation endpoint.
- Step 9 – Display the generated blog response.
- Step 10 – Send blog content to the summarization endpoint.
- Step 11 – Display the generated summary.
- Step 12 – Demonstrate comments, moderation, or other supporting functionality.

4. AI Demonstration

The central demonstration should show that AI is integrated into the backend workflow rather than functioning as an isolated interface. The BlogGenerator service connects to the AI Flow layer, while dedicated generation and summarization endpoints expose those capabilities to API clients.

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5. Security Demonstration

- Show that protected endpoints require authentication.
- Show that different roles receive different permissions.
- Demonstrate rejection of invalid credentials or unauthorized actions.
- Demonstrate validation of malformed input.
- Explain how passwords are hashed and how JWTs control stateless access.

6. Expected Demonstration Outcome

The final demonstration should establish that AI BlogNest operates as a functional, secure, modular backend for blog management and AI-assisted content creation. The evidence should connect the original requirements to implemented endpoints and visible API responses.

7. Final Submission Evidence

- Working backend project/code.
- API testing collection or screenshots.
- Registration and login evidence.
- Blog creation and retrieval evidence.
- AI blog-generation evidence.
- AI summarization evidence.
- Demo video.
- API testing video.
- Code repository/reference.